DATA SCIENCE · MIDTERM PROJECT REVIEW

Cross-Script Sentiment Transfer for Tunisian Dialect

Can a model that learned from Arabizi read Arabic script?
Business understanding through feature engineering

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WHERE WE ARE

The midway covers the first three CRISP-DM phases

PHASE 1

Business understanding

The problem, the research question, measurable success criteria

PHASE 2

Data understanding

Two sources, the variables, exploratory analysis and hypothesis tests

PHASE 3

Data preparation

Cleaning, merging, the cross-script mapping, the feature set

AFTER THE MIDWAY

Modeling and evaluation

Three models, with and without the mapping, on a learning curve

THE PROBLEM

Tunisians write one dialect in two scripts that share no characters

Arabic script

عيشك

“thank you”

=

Arabizi (Latin letters + digits)

3aychek

3 stands for ع

To a computer these are **two unrelated strings**: not one character in common.

Labeled Arabizi data is **plentiful**; labeled Arabic-script data is **scarce** and costly to annotate.

THE EVIDENCE

A model trained on one script fails on the other

Train on	Test on	Macro-F1
Arabic	Arabic	0.95
Arabizi	Arabizi	0.95
Arabic	Arabizi	0.63
Arabizi	Arabic	0.45

0.45

macro-F1 when an Arabizi-trained model reads Arabic script, worse than a coin flip on balanced classes (0.50)

Preliminary baseline from our proposal: TF-IDF + logistic regression on deduplicated TSAC, untuned, single run

RESEARCH QUESTION

Given a large labeled Arabizi corpus and a small labeled Arabic-script corpus, does **mapping both scripts to a common representation** improve sentiment classification on Arabic-script text, and **how does the gain vary** with the number of Arabic-script training examples?

Q1 · Does the mapping help?
Higher macro-F1 on Arabic-script test comments with no Arabic training data (k = 0)?

Q2 · How much Arabic data does it save?
How the gain changes as we add k = 100, 500, 1,000 and all Arabic examples

Q3 · Which model gains most?
TF-IDF + logistic regression, Naive Bayes, or a character CNN/BiLSTM

Q4 · Is the conversion accurate?
How close the automatic conversion is to one written by hand

OBJECTIVES AND SUCCESS CRITERIA

Success is defined in numbers before any model is trained

Business objective

Sentiment analysis of Arabic-script Tunisian without costly new labeling

Data science objective

A cleaning pipeline, a conversion module, three models compared with and without it

Criterion	Target	Status at midway
Mapping gain	+0.10 macro-F1 at k = 0 vs the same model without mapping	after modeling
Robustness	Gain larger than the spread across 5 seeds	after modeling
Small-data value	With mapping, k = 500 ≥ k = 1,000 without it	after modeling
Sanity floor	Beat “always negative” on the Arabic test set (57.5%)	floor measured
Conversion quality	Character error rate ≤ 0.15 against hand conversions	met: 0.105

DATA SOURCES

Two labeled corpora: one large Arabizi, one with both scripts

	TUNIZI V2	TSAC
Size	100,000 comments	17,069 (13,669 train / 3,400 test)
Script	Arabizi only	54% Arabizi, 44% Arabic, 1% mixed
Labels	53% pos, 44% neg, 3.5% neutral	52% pos, 48% neg
Origin	Social media (Fourati et al., 2021)	Facebook pages of Tunisian media, 2015–2016
Licence	MIT	LGPL-3.0
Role	Main Arabizi training data	Only Arabic-script source; the test set

Not used: the **neutral class** (3.5%, mostly off-topic) and **TUNIZI V1** (6,194 of its 9,210 sentences are already in V2). TSAC holds both scripts from the same pages, which isolates the effect of the script itself.

Sources: github.com/iCompass-ai/TUNIZI · github.com/fbougares/TSAC

HOW THE DATA IS USED

Train on Arabizi plus k Arabic comments, test on Arabic only

TRAINING SET

96,399
TUNIZI V2 · Arabizi

+

6,437
TSAC train · Arabizi

+

k
TSAC train · Arabic, k = 0, 100, 500, 1,000 or all 5,506

TEST SET

1,482
TSAC test · Arabic script

Never used for training or tuning. k = 0 is the hardest case: the model sees **no Arabic script at all**. Each setting is run with and without the mapping.

VARIABLES

One master table: four raw columns, six derived, two converted

Column	Type	Meaning	Origin
id	text	Stable id set on load, e.g. tsac_test_00001	raw
text	text	The comment after cleaning	raw
label	category	pos or neg (the target)	raw
source · split	category	tunizi / tsac · train / test	raw
script	category	arabic or arabizi (80% letter rule)	derived
n_chars · n_words	integer	Length in characters and in words	derived
has_emoji	boolean	At least one emoji	derived
has_arabizi_digits	boolean	A digit used as a letter (3aychek, not 2016)	derived
has_elongation	boolean	A character repeated 3+ times (ya3333jbni)	derived
text_norm · text_conv	text	Normalized text; normalized + converted	conversion

111,531 rows · no missing values after cleaning

CLEANING, EACH SOURCE ON ITS OWN

TUNIZI was nearly clean; TSAC lost 11%, mostly duplicates

Step	TUNIZI V2	TSAC
Loaded	100,000	17,069
Neutral class dropped	−3,522	none
Empty or without any letter	0	−75
Over 200 words (outliers up to 1,721)	−32	not cut
Same text, opposite labels: all copies	0	−107
Duplicates (keep one)	−2	−1,374
Train copies of test comments	none	−312
Cleaned	96,444	15,201

Merge: −45 TUNIZI copies of TSAC comments, −69 mixed-script comments → **111,531 comments**. Cleaning only removes rows and fixes encoding; it never rewrites words.

INTEGRATION AND LEAKAGE

Every treatment is chosen to keep the test set clean

Leakage removed

312 training comments also in the test set: dropped from train, so the official test set stays intact

Cross-dataset copies

45 TUNIZI comments also in TSAC, 14 of them TSAC test comments: the TUNIZI copy is dropped

Conflicting labels

12 texts appear as both positive and negative: all copies dropped, the true label is unknowable

Encoding repaired

260 _x000D_ artifacts, invisible characters, old Arabic letter forms: 16 real comments saved from deletion

Script detected

80% or more of the letters in one script; 69 mixed comments dropped; rule correct on 30/30 hand-checked rows

Fitted on train only

Vocabularies, scaling, feature selection and length cutoffs learn from training rows only

THE FINAL DATASET

111,531 comments, 94% Arabizi; the Arabic test set is 1,482

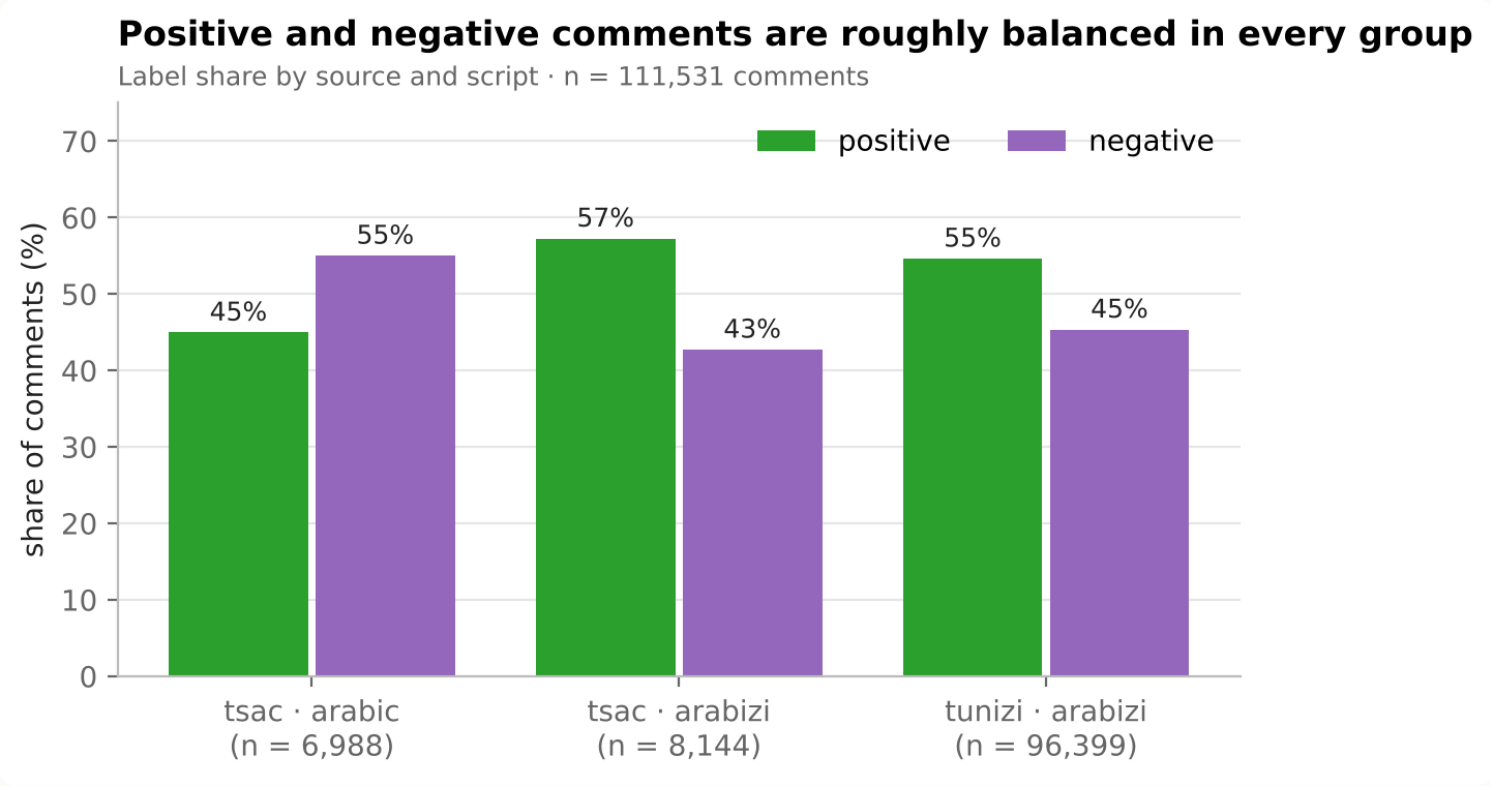
Source · split	Arabic	Arabizi	Total
TUNIZI · train	0	96,399	96,399
TSAC · train	5,506	6,437	11,943
TSAC · test	1,482	1,707	3,189
Total	6,988	104,543	111,531

1 : 15

Arabic-script vs Arabizi comments. That imbalance is why reusing the Arabizi data matters.

EDA · CLASS BALANCE

Classes are close to balanced in every group



What we see

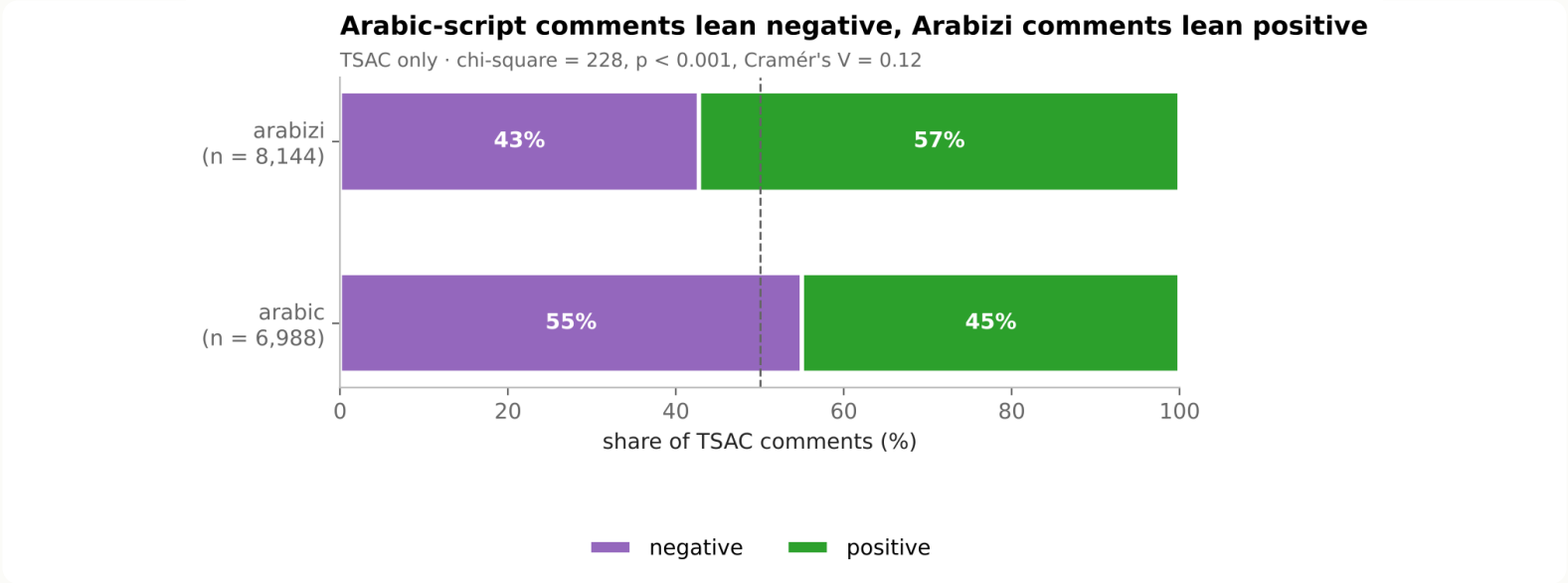
No group falls below 43% for either class, but the balance shifts with the script

So what

Main metric: **macro-F1**, which weighs both classes equally and does not reward guessing the larger class

EDA · HYPOTHESIS TEST 1: CHI-SQUARE OF INDEPENDENCE

Sentiment depends on the script, but weakly



Why this test

Two categorical variables in a 2 × 2 table; smallest expected count 3,383, far above 5

Result

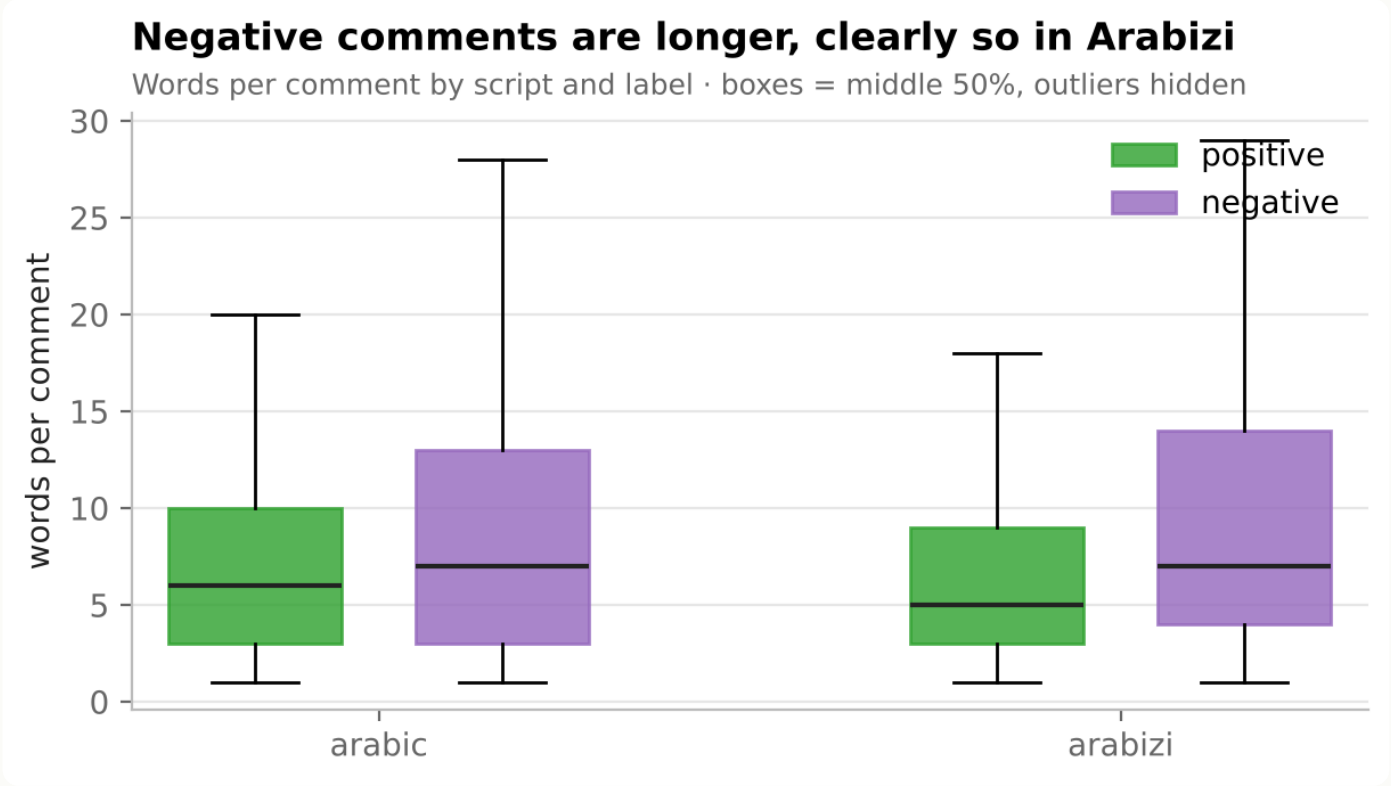
$\chi^2 = 227.5$, 1 df, $p \approx 2 \times 10^{-51}$: H0 rejected. Cramér's V = 0.12: a weak link

So what

Report every result per script. "Always negative" already scores 57.5% on the Arabic test set

EDA · HYPOTHESIS TEST 2: MANN-WHITNEY U

Negative comments are longer, clearly so in Arabizi



Why this test

Lengths are heavily skewed, not normal, so a rank test instead of a t-test

Result

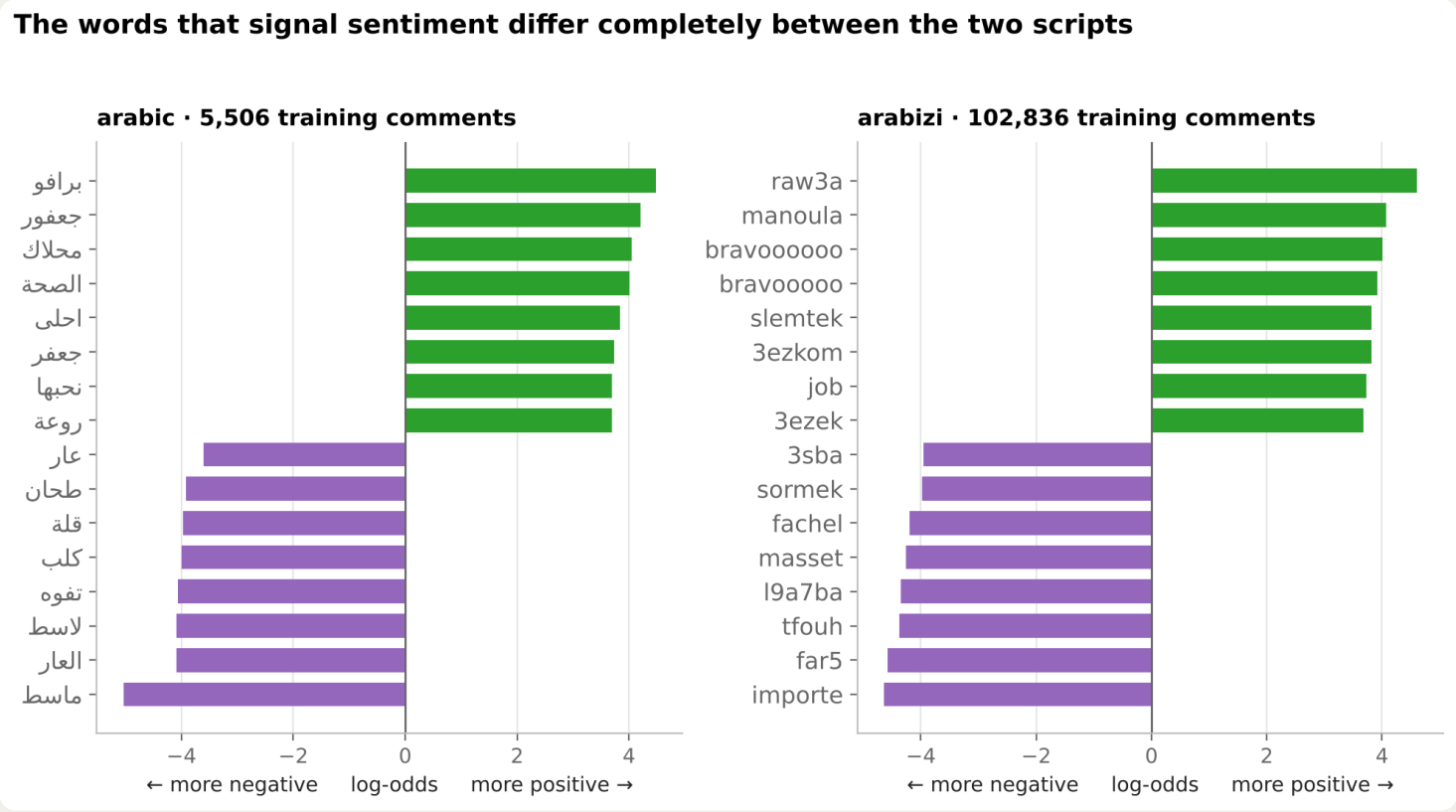
H0 rejected in both scripts. Effect size: **0.25 in Arabizi** (small), 0.08 in Arabic (negligible)

So what

Length is a weak extra signal; half the comments have 6 words or fewer

EDA · WORDS THAT CARRY SENTIMENT

The same sentiment words exist in both scripts, with no letter in common



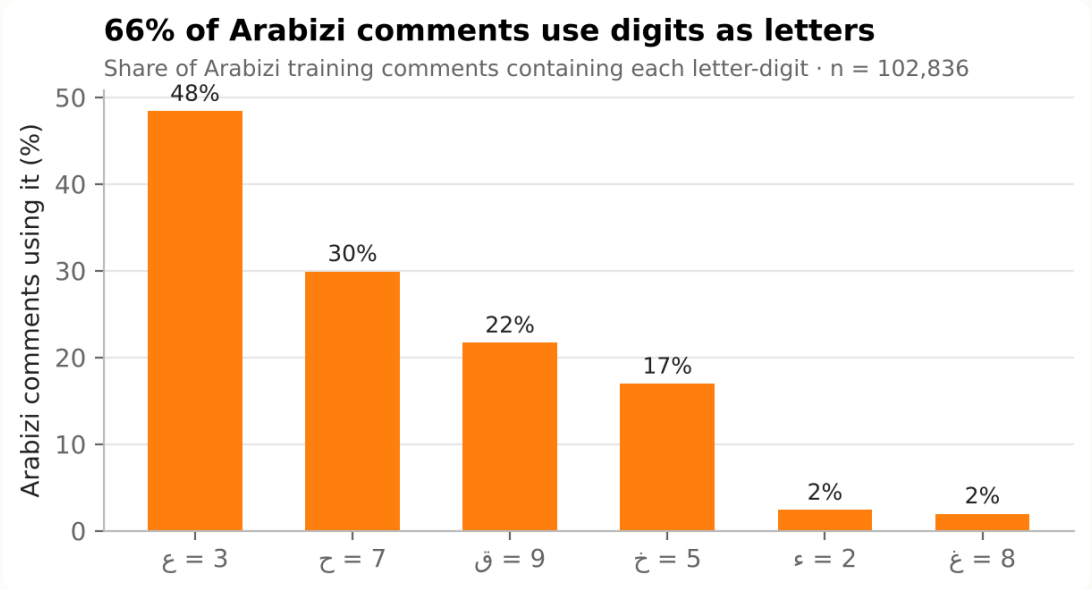
Ranked by **log-odds** between classes, not raw counts, to skip words used by both sides

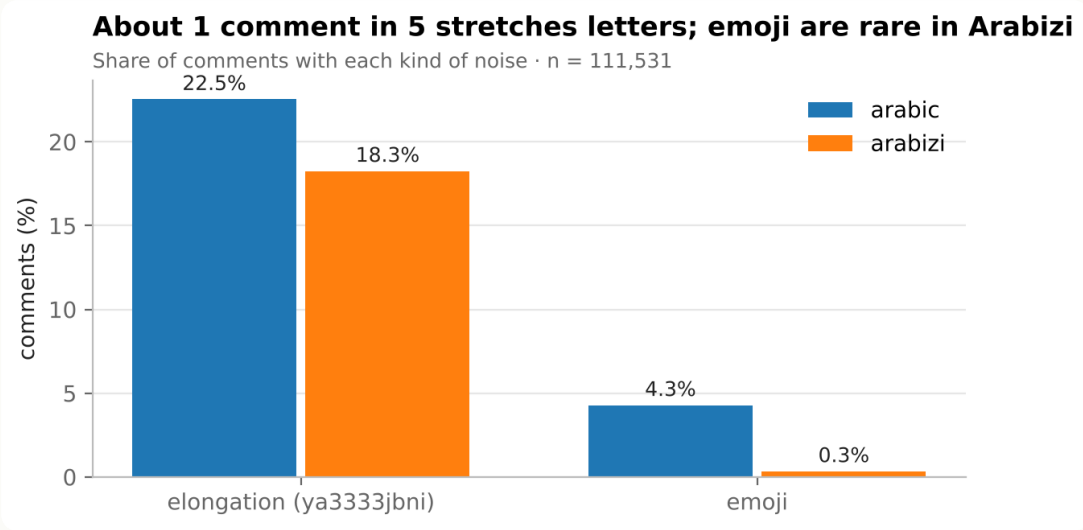
Arabic	Arabizi
ماسط	masset
تفوه	tfouh
روعة	raw3a

To a model these are different words: the gap the mapping must close

EDA · WRITING HABITS

Digits act as letters, and one comment in five stretches words



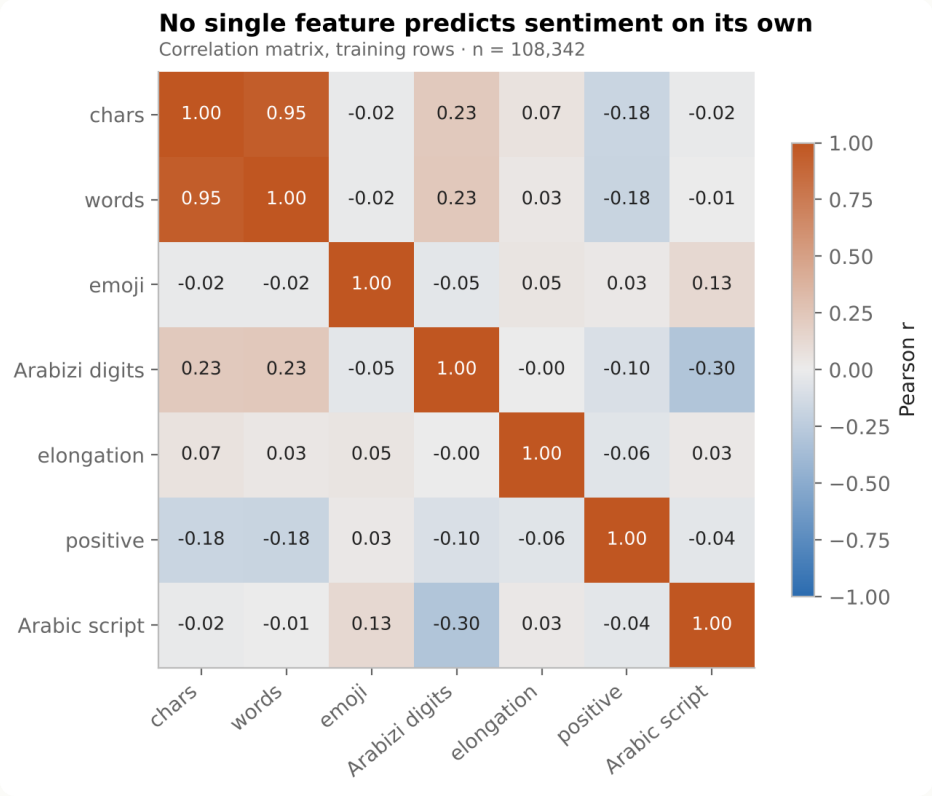


66% of Arabizi comments use 3, 7, 9 or 5 as letters → keep the digits; map خ ق ح ع onto them

~20% stretch letters (bravooooo) → collapse repeats. Emoji: 4.3% of TSAC, **0% of TUNIZI** → a limitation

EDA · CORRELATIONS

No summary feature predicts sentiment on its own



Pearson on numeric and 0/1 columns (= point-biserial and phi), training rows only

Pair	r	Decision
chars · words	0.95	keep one
length · positive	−0.18	minor feature
Arabizi digits · Arabic script	−0.30	drop: proxy for script
emoji, elongation · positive	≤ 0.06	weak

The signal is in the words, so the main features are character n-grams of the text

FEATURE ENGINEERING · STEP 1: NORMALIZATION

Every normalization step answers an EDA finding

Step	Example	EDA finding behind it
Lowercase, strip French accents	Très → tres	French mixed into Arabizi
Arabic: drop diacritics, unify alef forms	أحبك→ احبك	Same word, several spellings
Collapse 3+ repeated characters	ya3333jbni → ya3jbni	About 20% of comments
Keep digits used as letters	3aychek stays	66% of Arabizi comments
Space out emoji, drop skin tones	top 👍🏽 ! → top 👍 !	4.3% of TSAC comments

Normalization is applied in **both** conditions, with and without the mapping, so the comparison isolates the mapping itself

FEATURE ENGINEERING · STEP 2: THE CROSS-SCRIPT MAPPING

Both scripts become one vowel-free Latin form

ع → 3	ح → 7	خ → 5
ق → 9	ء → 2	ش → ch
غ → gh	ث → th	ض ظ ذ → dh

ا و ي ى ة → **removed (vowels)**

Other consonants keep their usual letter (b t d r z s f k l m n h).

Arabizi side: sh → ch, kh → 5, 8 → gh; then **all vowels removed** (a e i o u w y) and repeated letters collapsed

WORKED EXAMPLE

عيشك → **3chk**

3aychek → **3chk**

Arabic leaves short vowels out; Arabizi spells them freely (3aychek, 3ichek). Removing them on both sides makes the two meet.

FIRST EVIDENCE · VOCABULARY OVERLAP

After the mapping, an Arabizi vocabulary recognizes Arabic comments

before **2.1%** → after **91.9%**

of the words in Arabic-script test comments also appear in the Arabizi training vocabulary

1.5% → 73.1%
distinct test words found

1.4% → 32.5%
character 3-gram overlap (Jaccard)

88.4%
coverage counting only words of 3+ letters

Vocabularies built from training rows only · 1,482 Arabic test comments · 102,836 Arabizi training comments

Q4 · IS THE CONVERSION ACCURATE?

The automatic conversion stays close to a human’s: error rate 0.105

Direction	Error rate	Exact words	Rows
Arabic → Arabizi	0.107	62%	43
Arabizi → Arabic	0.101	68%	15
Overall	0.105	64%	58
Variant: keep w and y	0.196	39%	58

Target ≤ 0.15: **met**. Keeping w and y nearly doubles the error, which confirms removing all vowels.

Error rate = edit distance ÷ length of the converted human version. 60 comments hand-converted by the team; 2 excluded (words censored by the annotator).

What the remaining errors are

Word boundaries
Arabic attaches ال; writers type “el 7ounj” apart

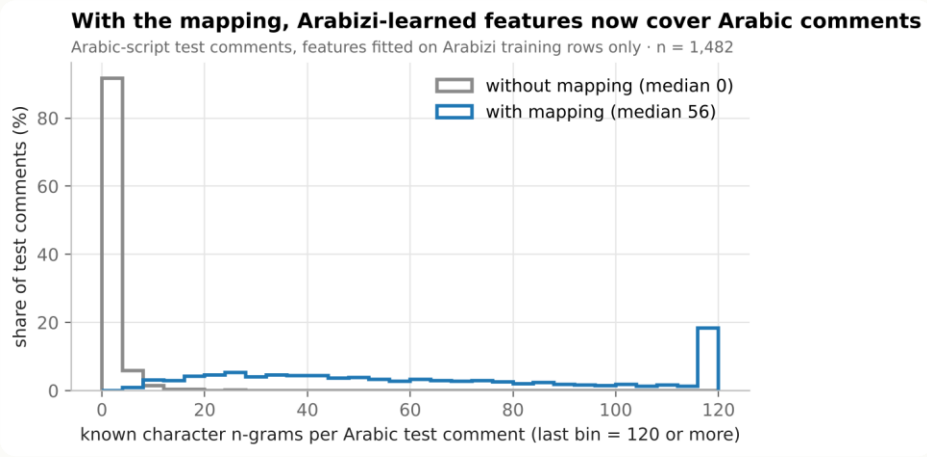
Spelling habits
h, k, th where the mapping gives 7, 9, dh

Annotator slips
A dropped letter or emoji in the hand version

FEATURE ENGINEERING · STEP 3: THE FINAL FEATURE SET

Character n-grams, fitted on training rows only

Feature	Why
TF-IDF char 2–5-grams, top 20,000 by chi-square	Robust to spelling variation; frequency filters then chi-square on training labels
log(1 + length), standardized	Weak link to sentiment (r = −0.18); log removes the skew (10.5 → 0.6)
Elongation flag (0/1)	Keeps the signal normalization removes
Character sequences (CNN/BiLSTM)	Cut at the 95th percentile of training lengths
Excluded: script, digit flag, emoji, word count	Script shortcut (chi-square), absent from TUNIZI, duplicate of length



71%

of Arabic test comments invisible without mapping

0%

invisible with mapping (median 56 known n-grams)

CONCLUSIONS

The data is ready, and the mapping closes the vocabulary gap

Question	Evidence so far	Status
Q1 · Does the mapping help?	Test words known to Arabizi training: 2.1% → 91.9%; invisible comments 71% → 0%	promising; models next
Q2 · Arabic data saved?	Learning curve designed: k = 0 to 5,506	after modeling
Q3 · Which model?	Features ready for all three models, both conditions	after modeling
Q4 · Conversion accurate?	Error rate 0.105 vs a human (target ≤ 0.15)	answered

Data
111,531 clean comments, no leakage, test set of 1,482

EDA
Script–sentiment link is real but weak: report per script, floor 57.5%

Preparation
Same words, no shared letters, until the mapping makes them meet

LIMITATIONS

What these results cannot tell us yet

- ⚠ **Overlap is not accuracy.** Shared vocabulary does not prove better sentiment predictions
- ⚠ **Small manual sample.** 58 scored comments, fewer than the 150–200 planned
- ⚠ **French and emoji.** Code-switching is not mapped; TUNIZI has no emoji at all
- ⚠ **Collisions.** Without vowels, some different words share a skeleton (7ob, 7abb → 7b)
- ⚠ **One source, one period.** Arabic script only from TSAC, Facebook media pages 2015–2016
- ⚠ **Short mixed comments.** One Latin word can push a short comment below 80% and drop it

NEXT STEPS · MODELING AND EVALUATION

One experiment answers Q1 to Q3: 150 training runs

3
models: TF-IDF + LR, Naive Bayes, char CNN/BiLSTM

×

2
conditions: with and without the mapping

×

5
values of k: 0, 100, 500, 1,000, all

×

5
random seeds, reported as mean ± std

Evaluate

Macro-F1, accuracy and confusion matrices per script, against the 57.5% floor and our success criteria

Analyze errors

Sarcasm, negation, French code-switching, conversion failures

Improve the mapping

Version 2: attached ال, the Tunisian letter ڤ, repeated digits

THANK YOU

Questions?